



IIT JEE | MEDICAL | FOUNDATION

illuminati - 2020

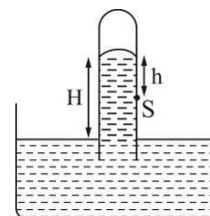
Advanced Physics Assignment-2

Liquids, Properties of Matter, Thermodynamics, SHM & Wave Motion

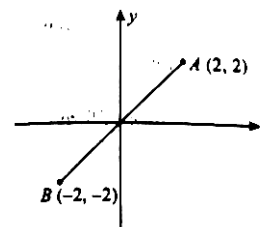
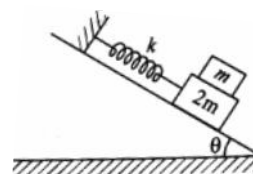
SECTION-1

SINGLE CORRECT ANSWER TYPE

- A vertical capillary is brought in contact with water surface. The radius of the capillary is r , the angle of contact is θ and surface tension of water is σ . The heat liberated while the water rises in the capillary is:
 (A) $\frac{\sigma^2 \cos^2 \theta}{\rho g}$ (B) $\frac{\pi \sigma^2}{\rho g \cos^2 \theta}$ (C) $\frac{2\pi \sigma^2 \cos^2 \theta}{\rho g}$ (D) $\frac{2\pi \sigma^2}{\rho g \cos \theta}$
- A liquid of volumetric thermal expansion coefficient γ and bulk modulus B is filled in a spherical tank of negligible heat expansion coefficient. Its radius is R and wall thickness is t ($t \ll R$). When the temperature of the liquid is raised by θ , the tensile stress developed in the walls of the tank is:
 (A) $B\gamma \theta R / 2t$ (B) $B\gamma \theta R / t$ (C) $2B\gamma \theta R / t$ (D) $B\gamma \theta R / 4t$
- Which of the following is greatest in SHM ? (Assuming potential energy = 0 at mean position)
 (A) Average kinetic energy with respect to space
 (B) Average potential energy with respect to space
 (C) Average kinetic energy with respect to time
 (D) Average potential energy with respect to time
- A solid sphere and a hollow sphere of the same material and size are heated to the same temperature and allowed to cool in the same surroundings. If the temperature difference between the surroundings and each sphere is T , then:
 (A) The hollow sphere will cool at a faster rate for all values of T
 (B) The solid sphere will cool at a faster rate for all values of T
 (C) Both spheres will cool at the same rate for all values of T
 (D) Both spheres will cool at the same rate only for small values of T
- What is the fraction of molecule below an altitude h in atmosphere ? Assume uniform gravitational field, isothermal conditions, mass of a molecule m , Boltzman constant k , temperature T .
 (A) $f = e^{(mgh/kT)}$ (B) $f = e^{-(mgh/kT)}$ (C) $f = 1 - e^{-(mgh/kT)}$ (D) $f = 1 - e^{(mgh/kT)}$
- In a mercury barometer used to measure atmospheric pressure, mercury rises to a height H . Now if a small hole is made at point S as shown, mercury comes out from this hole with a speed equal to :
 (A) $\sqrt{2gh}$ (B) $\sqrt{2g(H-h)}$
 (C) $\sqrt{2gH}$ (D) None of these



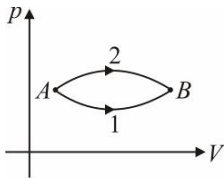
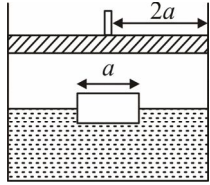
7. Two spheres of same metal have the same volume. But one is solid and the other is hollow. When the change in temperature of both of them is same, which of the following statements about the change in their diameters due to thermal expansion is true?
 (A) More for solid sphere (B) More for hollow sphere
 (C) Same for both spheres (D) It cannot be predicted
8. Two waves are propagating to the point P along a straight line produced by two sources A and B of simple harmonic sound of equal frequency. The amplitude of each wave at P is ' a ' and the phase of A is ahead by $\pi/3$ than that of B and the distance AP is greater than BP by 50 cm. Then the resultant amplitude at the point P will be, if the wavelength is 1 meter:
 (A) $2a$ (B) $a\sqrt{3}$ (C) $a\sqrt{2}$ (D) a
9. An accurate pendulum clock is mounted on the ground floor of a high building. How much time will it lose or gain in one day if it is transferred to top storey of a building which is $h = 200$ m higher than the ground floor. Radius of earth is 6.4×10^6 m :
 (A) It will lose 6.2 s (B) It will lose 2.7 s (C) It will gain 5.2 s (D) It will gain 1.6 s
10. The coefficient of friction between block of mass m and $2m$ is $\mu = 2 \tan \theta$. There is no friction between block of mass $2m$ and inclined plane. The maximum amplitude of two block system for which there is no relative motion between both the blocks is :
 (A) $g \sin \theta \sqrt{\frac{k}{m}}$ (B) $\frac{mg \sin \theta}{k}$ (C) $\frac{3mg \sin \theta}{k}$ (D) None of these
11. A vertical glass jar is filled with water at 10°C . It has one thermometer at the top and another at the bottom. The central region of the jar is gradually cooled. It is found that the bottom thermometer reads 4°C earlier than the top thermometer. And the top thermometer reads 0°C earlier than the bottom thermometer. This happens because :
 (A) The top thermometer is faulty (B) The bottom thermometer is faulty
 (C) Density of water is maximum at 4°C (D) Density of water is minimum at 0°C
12. Two identical heaters are coated with paint. In 1st case $e_1 = 1.0$ and in 2nd case $e_2 = 0.5$. Both are kept in identical chambers which are in similar outside surroundings. If the heaters are switched on, in steady state 1st heater has temperature T_1 on surface and θ_1 of its chamber. 2nd heater has temperature T_2 on surface and θ_2 of its chamber. Then :
 (A) $\theta_1 = \theta_2$; $T_1 < T_2$ (B) $\theta_1 > \theta_2$; $T_1 = T_2$ (C) $\theta_1 = \theta_2$; $T_1 > T_2$ (D) $\theta_1 < \theta_2$; $T_1 = T_2$
13. A particle of mass $m = 2$ kg executes SHM in xy -plane between points A and B under action of force $\vec{F} = F_x \hat{i} + F_y \hat{j}$. Minimum time taken by particle to move from A to B is 1 sec. At $t = 0$ the particle is at $x = 2$ and $y = 2$. Then F_x as function of time t is:
 (A) $-4\pi^2 \sin \pi t$ (B) $-4\pi^2 \cos \pi t$
 (C) $4\pi^2 \cos \pi t$ (D) None of these



14. The wavelength of light of a particular wavelength received from a galaxy is measured on earth and is found to be 0.5% more than its wavelength. It follows that the galaxy is:
- (A) Approaching the earth with a speed $3 \times 10^6 \text{ ms}^{-1}$
 (B) Going away from the earth with a speed $1.5 \times 10^6 \text{ ms}^{-1}$
 (C) Approaching the earth with a speed $1.5 \times 10^6 \text{ ms}^{-1}$
 (D) Going away from the earth with a speed $1.5 \times 10^6 \text{ ms}^{-1}$
15. A planet contains a single type of gas in its atmosphere having molecular mass, $m = 3.0 \times 10^{-26} \text{ kg}$. Distribution of speeds in atmosphere is

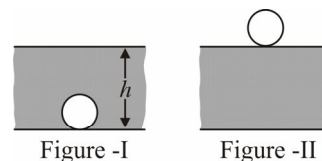
Speed (ms^{-1})	Percentage of molecules
200	10
400	20
600	40
800	20
1000	10

If escape velocity of the planet is 900 ms^{-1} , assuming other conditions are stable. The possible estimated reduction in temperature of the planet in few years will be :

- (A) 100 K (B) 96 K (C) 296 K (D) 48 K
16. A gas undergoes a change of state from A to B via two paths (path 1 and path 2).
 Ratio of molar heat capacities in path 1 and path 2 is:
- (A) < 1 (B) > 1
 (C) 1 (D) Data insufficient
- 
17. A smooth uniform string of natural length L_0 , cross-sectional area A and Young's modulus Y is pulled along its length by a force F on a horizontal smooth surface. The elastic potential energy stored in the string is
- (A) $\frac{2F^2 L_0}{AY}$ (B) $\frac{F^2 L_0}{3AY}$ (C) $\frac{F^2 L_0}{2AY}$ (D) $\frac{F^2 L_0}{6AY}$
18. Two adiabatic vessels, each containing the same mass m of water but at different temperatures, are connected by a rod of length L , cross-section A , and thermal conductivity K . The ends of the rod are inserted into the vessels, while the rest of the rod is insulated so that there is negligible loss of heat into the atmosphere. The specific heat capacity of water is s , while that of the rod is negligible. The temperature difference between the two vessels reduces to $1/e$ of its original value after a time, Δt . The thermal conductivity (K) of the rod may be expressed by:
- (A) $\frac{msL}{A\Delta t}$ (B) $\frac{emsL}{A\Delta t}$ (C) $\frac{msL}{2eA\Delta t}$ (D) $\frac{msL}{2A\Delta t}$
19. A cubical block of side a is floating in a fixed and closed cylindrical container of radius $2a$ kept on the ground. Density of the block is ρ , whereas density of the liquid is 2ρ . Container is made up of conducting wall, so that the temperature remains constant. A piston is mounted in the cylinder which can move inside the cylinder without friction. If piston oscillates with a large amplitude A , then
- 

- (A) the cube will remain stationary
 (B) the cube will oscillate with very small amplitude in opposite phase with piston
 (C) the cube will oscillate with very small amplitude in same phase with piston
 (D) the cube will oscillate with amplitude A

20. A sphere of mass m and radius r rests at the bottom of a large reservoir of water as shown in figure-I. Depth of the reservoir is h . Density of material of the sphere is the same as that of water. Now the sphere is slowly pulled completely out of water as shown in figure-II. Which of the following is a correct expression for work done by the agent pulling the sphere?



- (A) mgr (B) $0.5 mgr$ (C) $mg(r + h)$ (D) $mg(0.5r + h)$

SECTION-2

LINK-COMPREHENSION TYPE

PARAGRAPH FOR QUESTIONS 21 - 23

Solid and liquid both expands on heating. The density of substance decreases on expanding according to the relation

$$\rho_2 = \frac{\rho_1}{1 + \gamma(T_2 - T_1)}$$

where,

$\rho_1 \rightarrow$ density at T_1

$\rho_2 \rightarrow$ density at T_2

$\gamma \rightarrow$ coefficient of volume expansion of substances

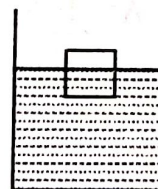
When a solid is submerged in a liquid, liquid exerts an upward force on solid which is equal to the weight of liquid displaced by submerged part of solid. Solid will float or sink depends on relative densities of solid and liquid. A cubical block of solid floats in a liquid with half of its volume submerged in liquid as shown in figure (at temperature T).

$\alpha_s \rightarrow$ coefficient of linear expansion of solid

$\gamma_L \rightarrow$ coefficient of volume expansion of liquid

$\rho_s \rightarrow$ density of solid at temperature T

$\rho_L \rightarrow$ density of liquid at temperature T



21. If fraction submerged does not change on increasing temperature the relation between γ_L and α_s is:

- (A) $\gamma_L = 3\alpha_s$ (B) $\gamma_L = 2\alpha_s$ (C) $\gamma_L = 4\alpha_s$ (D) $\gamma_L = \left(\frac{3}{2}\right)\alpha_s$

22. If the depth of the block submerged in the liquid does not change on increasing temperature then :

- (A) $\gamma_L = 2\alpha_s$ (B) $\gamma_L = 3\alpha_s$ (C) $\gamma_L = \left(\frac{3}{2}\right)\alpha_s$ (D) $\gamma_L = \left(\frac{4}{3}\right)\alpha_s$

23. Assume block does not expand on heating. The temperature at which the block just begins to sink in liquid is :

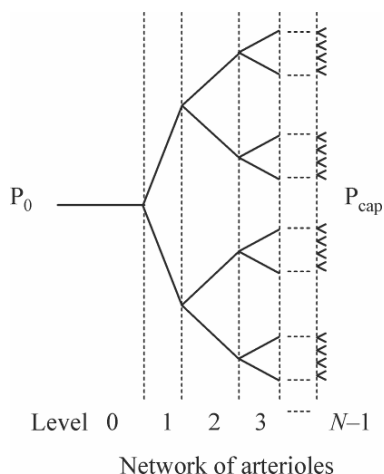
- (A) $T + \frac{1}{\gamma_L}$ (B) $T + \frac{1}{(2\gamma_L)}$ (C) $T + \frac{2}{\gamma_L}$ (D) $T + \frac{\gamma_L}{2}$

PARAGRAPH FOR QUESTIONS 24 - 25

Blood vessels are approximately cylindrical in shape, and it is known that for a steady, non-turbulent flow of an incompressible fluid in a rigid cylinder, the difference in pressure of the fluid at the two ends of the cylinder is given by: $\Delta P = \frac{8\ell\eta}{\pi r^4} Q$

Where ℓ and r are the length and radius of the cylinder, η is the fluid viscosity and Q is the volumetric flow rate, i.e. the fluid volume that passes the cylinder cross-section per unit time. This expression is often able to provide the correct order of magnitude for the pressure difference in a blood vessel. Moreover, this expression has the same form as Ohm's law, with the volumetric flow rate being interpreted as a current, the difference in pressure as a voltage, and the factor $R = \frac{8\ell\eta}{\pi r^4}$ as a resistance.

Consider for example the symmetrical network of arterioles (small arteries) depicted in figure that delivers blood to the capillary bed of a tissue. In this network, at each bifurcation a vessel is divided in two identical vessels. However, the vessels of higher levels are thinner and shorter : Considered that the radii and lengths of vessels in two consecutive levels i and $i + 1$, are related by $r_{i+1} = r_i / 2^{1/3}$ and $\ell_{i+1} = \ell_i / 2^{1/3}$.



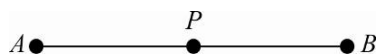
24. Obtain an expression for the volumetric flow rate, Q_i , in a vessel at any level i , as a function of the total number of levels N , of the viscosity η , of the radius r_0 and length ℓ_0 of the arteriole at level 0, and of the difference $\Delta P = P_0 - P_{cap}$ between the inlet pressure at the arteriole at level 0 (P_0) and the pressure at the capillary bed (P_{cap}).

(A) $\Delta P \frac{\pi r_0^4}{8N \ell_0 \eta}$ (B) $\Delta P \frac{\pi r_0^4}{2^{i+3} N \ell_0 \eta}$ (C) $\Delta P \frac{\pi r_0^4}{2^{i+3} \ell_0 \eta}$ (D) $\Delta P \frac{\pi r_0^4}{2^i N \ell_0 \eta}$

25. Calculate the numerical value of the volumetric flow rate Q_0 of the arteriole at level 0, if its radius is $6.0 \times 10^{-5} \text{ m}$ and its length is $2.0 \times 10^{-3} \text{ m}$. Consider that the pressure at the arteriole inlet is 55 mmHg and the vessel network has $N = 6$ levels linking this arteriole to the capillary bed at the pressure 30 mmHg . Consider that the blood viscosity is $\eta = 3.5 \times 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$. Express your result in mL/h . (Normal atmospheric pressure, $P_0 = 1.013 \times 10^5 \text{ Pa} = 760 \text{ mmHg}$)
- (A) 15 mL/hr (B) 4 mL/hr (C) 1.5 mL/hr (D) 150 mL/hr

PARAGRAPH FOR QUESTIONS 26 - 28

A and B are two fixed points at a distance 3ℓ apart. A particle of mass m placed at a point P experiences the force $2(mg/\ell)\overline{PA}$ and the force $(mg/\ell)\overline{PB}$ simultaneously. Initially at $t = 0$, the particle is projected from A towards B with speed $3\sqrt{g\ell}$.



26. The particle moves simple harmonically with period T and amplitude A :

- (A) $A = 2\ell, T = 2\pi\sqrt{\frac{\ell}{g}}$ (B) $A = 3\ell, T = 2\pi\sqrt{\frac{\ell}{2g}}$
 (C) $A = 2\ell, T = 2\pi\sqrt{\frac{\ell}{3g}}$ (D) $A = \ell, T = 2\pi\sqrt{\frac{\ell}{g}}$

27. The instant 't' at which the particle arrives at B in terms of the periodic time T will be :

- (A) $t = \frac{T}{2}$ (B) $t = \frac{T}{3}$ (C) $t = \frac{T}{4}$ (D) $t = \frac{2T}{3}$

28. The velocity of the particle when it reaches B will be :

- (A) zero (B) $3\sqrt{g\ell}$ (C) $2\sqrt{g}$ (D) $g\ell$

PARAGRAPH FOR QUESTIONS 29 - 31

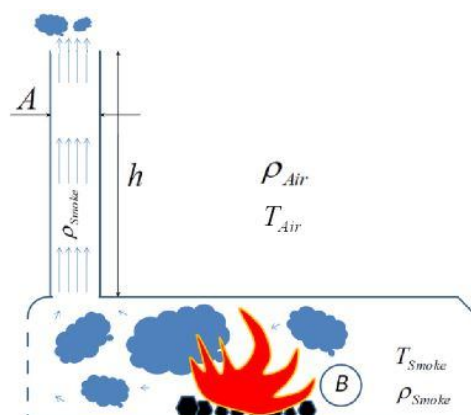
Gaseous product of burning are released into the atmosphere of temperature T_{Air} through a high chimney of cross-section A and height h (see figure). The solid matter is burned in the furnace which is at temperature T_{smoke} . The volume of gases produced per unit time in the furnace is B .

Assume that :

- The velocity of the gases in the furnace is negligibly small.
- The density of the gases (smoke) does not differ from that of the air at the same temperature and pressure; while in furnace, the gases can be treated as ideal.
- The pressure of the air changes with height in accordance with the hydrostatic law ; the change of the density of the air (ρ_{air}) with height is negligible. Pressure inside the furnace is same as outside.
- The flow of gases fulfills the Bernoulli equation which states that the following quantity is conserved in all points of the flow : $\frac{1}{2}\rho v^2(z) + \rho gz + p(z) = \text{const.}$

where ρ is the density of the gas, $v(z)$ is its velocity, $p(z)$ is pressure, and z is the height .

- The change of the density of the gas (ρ_{gas}) is negligible throughout the chimney.



29. What is the minimal height of the chimney needed in order that the chimney functions efficiently, so that it can release all of the produced gas into the atmosphere? Express your result in terms of $B, A, g, \rho_{air}, \rho_{smoke}$.

(A) $\frac{B^2 \rho_{smoke}}{2A^2 g (\rho_{air} - \rho_{smoke})}$ (B) $\frac{B^2 \rho_{air}}{2A^2 g (\rho_{air} - \rho_{smoke})}$

(C) $\frac{B^2}{2A^2 g}$ (D) $\frac{B^2}{2A^2 g} \frac{\rho_{smoke}}{\rho_{air}}$

30. Assume that two chimneys are built to serve exactly the same purpose. Their cross-sections are identical, but are designed to work in different parts of the world : One in cold regions, designed to work at an average atmospheric temperature of -30°C and the other in warm regions, designed to work at an average atmospheric temperature 30°C . The temperature of the furnace is 400°C . It was calculated that the height of the chimney designed to work in cold regions is 100m . How high is the other chimney?

(A) 116 m (B) 200 m (C) 145 m (D) 50 m

31. How does the pressure of the gases vary with height (z) inside the chimney? Here $P(0)$ is the pressure inside the furnace and h is the minimal height as calculated above.

(A) $P(z) = P(0) - \rho_{smoke} g z$

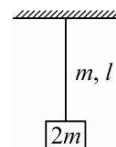
(B) $P(z) = P(0) - (\rho_{air} - \rho_{smoke}) g h - \rho_{smoke} g z$

(C) $P(z) = P(0) - (\rho_{air} - \rho_{smoke}) g z$

(D) $P(z) = P(0) - \rho_{air} g h - \rho_{smoke} g z$

PARAGRAPH FOR QUESTIONS 32 - 33

A block of mass $2m$ is hanging at the lower end of a rope of mass m and length l , the other end being fixed to the ceiling. A pulse of wavelength λ_0 is produced at the lower end of the rope.



32. The wavelength of the pulse when it reaches the other end of the rope is:

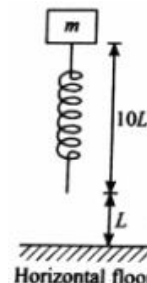
(A) $\sqrt{3}\lambda_0$ (B) $\sqrt{\frac{3}{2}}\lambda_0$ (C) λ_0 (D) $\frac{\lambda_0}{2}$

33. The time taken by the pulse to reach the other end of the rope is:

(A) $2\sqrt{\frac{l}{g}}(\sqrt{3}-1)$ (B) $2\sqrt{\frac{l}{g}}(\sqrt{3}-2)$ (C) $2\sqrt{\frac{l}{g}}$ (D) $2\sqrt{\frac{l}{g}}(\sqrt{3}-\sqrt{2})$

PARAGRAPH FOR QUESTIONS 34 - 35

A small block of mass m is fixed at upper end of a massless vertical spring of spring constant $K = \frac{4mg}{L}$ and natural length ' $10L$ '. The lower end of spring is free and is at a height L from fixed horizontal floor as shown. The spring is initially unstressed and the spring-block system is released from rest in the shown position.



34. As the block is coming down, the maximum speed attained by the block is :

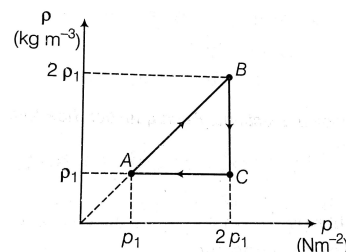
- (A) $\sqrt{gL}$ (B) $\sqrt{3gL}$ (C) $\frac{3}{2}\sqrt{gL}$ (D) $\sqrt{\frac{3}{2}gL}$

35. Till the block reaches its lowest position for the first time, the time duration for which the spring remains compressed is :

- (A) $\pi\sqrt{\frac{L}{2g}} + \sqrt{\frac{L}{4g}}\sin^{-1}\frac{1}{3}$ (B) $\frac{\pi}{4}\sqrt{\frac{L}{g}} + \sqrt{\frac{L}{4g}}\sin^{-1}\frac{1}{3}$
 (C) $\pi\sqrt{\frac{L}{2g}} + \sqrt{\frac{L}{4g}}\sin^{-1}\frac{2}{3}$ (D) $\frac{\pi}{2}\sqrt{\frac{L}{2g}} + \sqrt{\frac{L}{4g}}\sin^{-1}\frac{2}{3}$

PARAGRAPH FOR QUESTIONS 36 - 37

Consider a thermodynamic cycle as shown in which one mole of a monatomic ideal gas is taken from state A to B to C to A.



36. Ratio of work done by the gas in the process, AB, BC and CA ($W_{AB} : W_{BC} : W_{CA}$) is:

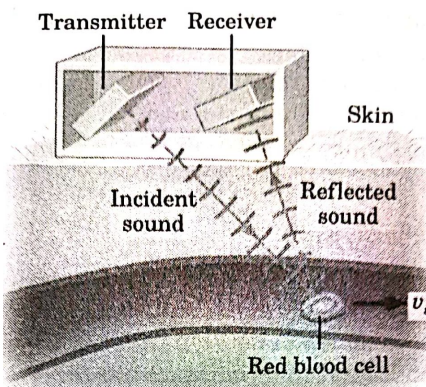
- (A) $\ln 2 : 1 : 1$ (B) $\ln 2 : 1 : 0$
 (C) $\ln 0.5 : 1 : 0$ (D) $\ln 0.5 : 1 : 1$

37. Ratio of heat rejected by gas to the total heat supplied is:

- (A) $\frac{2}{5}(1 - \ln 2)$ (B) $\frac{5}{2}(1 - \ln 2)$ (C) $\frac{2}{5}\ln 2$ (D) $\left(\frac{3 + \ln 4}{5}\right)$

PARAGRAPH FOR QUESTIONS 38 - 40

The Doppler flow meter is a particularly interesting medical application of the Doppler Effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the red blood cells, its frequency is changed in a kind of Doppler Effect because the cells are moving with the same velocity as the blood. The receiving element detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the speed of the blood flow can be determined. Typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.



38. Assume that the red blood cell is directly moving away from the source and the receiver. What is the (approx.) speeds of the sound wave in the blood ?

- (A) 1700 m/s (B) 330 m/s (C) 5000 m/s (D) 3000 m/s

39. An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency due to reflection from the red blood cell in that region ?
 (A) 150 Hz (B) 300 Hz (C) 600 Hz (D) 2400 Hz
40. At what extra rate does the heart have to work due to this narrowing down of the artery ? Assume the density to be 1.5 gm/cc and the area of the normal artery to be 0.1 cm^2 .
 (A) $1.125 \times 10^{-4} \text{ W}$ (B) $2.5 \times 10^{-4} \text{ W}$ (C) $6.25 \times 10^{-5} \text{ W}$ (D) $5.625 \times 10^{-5} \text{ W}$

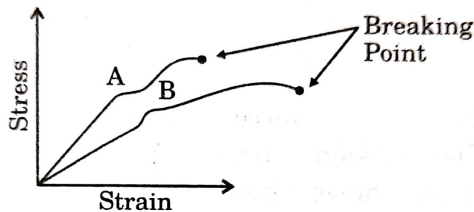
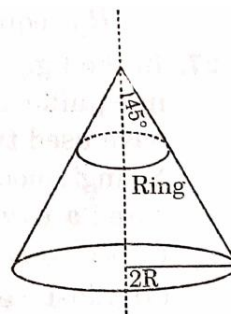
SECTION-3

MULTIPLE CORRECT ANSWER TYPE

41. A spherical object which behaves as a black body is raised to the temperature T_0 and placed in an environment, initially at 0°C . While the object cools from T_0 to the temperature T_e in an interval of time $t = \tau$, the environment rises to the temperature $t_0^\circ\text{C}$ during the same time. The average rate of cooling of the object, $\left(\frac{dT}{dt}\right)$ and the average net heat loss (Q) of the object during the time interval $0 \leq t \leq \frac{\tau}{4}$ are respectively given by $\left(\frac{dT}{dt}\right) = \alpha_A$ and $(Q) = \beta_A$ while the same quantities during the interval $\frac{3\tau}{4} \leq t \leq \tau$ are $\left(\frac{dT}{dt}\right) = \alpha_B$ and $(Q) = \beta_B$. Read the following statement about this system.

State which of the above statements are true, by correct choice from below :

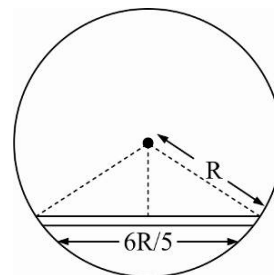
- (A) $\alpha_A > \alpha_B, \beta_A > \beta_B$ (B) $\alpha_A < \alpha_B, \beta_A < \beta_B$
 (C) The rate of cooling decreases continuously with time
 (D) The rate of cooling decreases with time at first, reaches a minimum and then increases with time
42. A ring of mass m , radius R , cross-sectional area A and Young's modulus Y is kept on a smooth cone of radius $2R$ and semi vertical angle 45° , as shown in the figure. Assume that the extension in the ring is small :
 (A) The tension in the ring will be same throughout
 (B) The tension in the ring will be independent of the radius of ring
 (C) The extension in the ring will be $\frac{mgR}{AY}$
 (D) Elastic potential energy stored in the ring will be $\frac{m^2 g^2 R}{8\pi YA}$
43. Select the correct statement on the basis of the given graph :



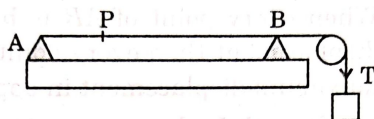
- (A) Young's modulus of A is greater than B (B) Young's modulus of A is less than B
 (C) A is less ductile than B (D) A is more ductile than B

44. A small squirrel is put into a circular wheel cage of radius R which has a frictionless central pivot. A horizontal platform of length $\frac{6R}{5}$ is fixed to the wheel below the pivot as shown. Initially squirrel is at rest at one end of the platform. When the platform is released squirrel starts running but platform and wheel remain stationary. Choose the correct options :

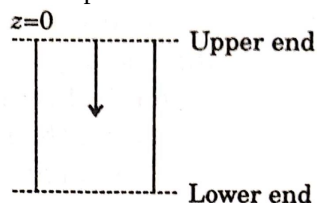
- (A) Maximum speed of squirrel is $\sqrt{\frac{Rg}{5}}$
 (B) Maximum speed of squirrel is $\sqrt{\frac{9Rg}{20}}$
 (C) Maximum acceleration of squirrel is $\frac{4g}{3}$
 (D) Maximum acceleration of squirrel is $\frac{3g}{4}$



45. A sonometer string AB of length 1m is stretched by a load and the tension T is adjusted so that the string resonates to a frequency of 1 kHz . Any point P of the wire may be held fixed by use of a movable bridge that can slide along the base of sonometer.



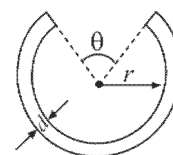
- (A) If point P is fixed so that $AP : PB :: 1 : 4$, the smallest frequency (fundamental) for which the sonometer wire resonates is 5 kHz
 (B) If P is taken at midpoint of AB and fixed, then, when the wire vibrates in the third harmonic of its fundamental, the number of nodes in the wire (including A and B) will be totally seven
 (C) If the fixed point P divides AB in the ratio $1 : 2$, then the tension needed to make the string vibrate at 1 kHz will be $3T$
 (D) The fundamental frequency of the sonometer wire when P divides AB in the ratio $a : b$ will be the same as the fundamental frequency when P divides AB in the ratio $b : a$
46. In an organic pipe (which may be closed or open) of 99 cm length, standing wave is set up whose equation is given by longitudinal displacement $x = 0.3\text{ mm} \cos\left[\left(\frac{2\pi}{0.8}\right)(z + 0.01)\right] \cos(400t)$, where z is measured from the top of tube in metres and t is in seconds.



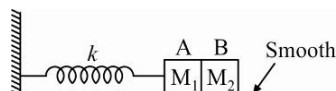
Take end correction = 1 cm . Choose correct statement(s).

- (A) The upper end and lower end of the tube are open and closed respectively
 (B) The upper end and lower end of tube are closed and open respectively
 (C) The air column is vibrating in third harmonic
 (D) The air column is vibrating in second overtone

47. A source is moving on a circle given by the equation $x^2 + y^2 = R^2$, with constant speed $\frac{330\pi}{6\sqrt{3}} m/s$, in anti-clockwise sense. A detector is at rest at point $(2R, 0)$ w.r.t. the centre of the circle. If the frequency emitted by the source is f and the speed of sound, $c = 330 m/s$. Then :
- (A) the position of the source when the detector records the maximum frequency is $\left(+\frac{\sqrt{3}}{2}R, -\frac{R}{2} \right)$
- (B) the coordinate of the source when the detector records minimum frequency is $(0, R)$
- (C) the minimum frequency recorded by the detector is $\frac{6\sqrt{3}}{\pi + 6\sqrt{3}} f$
- (D) the maximum frequency recorded by the detector is $\frac{6\sqrt{3}}{6\sqrt{3} - \pi} f$
48. Two holes with area $a = 0.2 \text{ cm}^2$, each are drilled one above the other in the wall of a vertical vessel filled with water. The distance between the holes is $H = 50 \text{ cm}$. Every second $Q = 140 \text{ cm}^3$ of water is poured into the vessel. The streams flowing out of the holes intersect at a point. x is the horizontal distance of this point from the vessel and y is the vertical distance of this point from the water surface in the tank. x and y remains same as the time passes. Choose the correct option(s).
- (A) $x = 120 \text{ cm}$ (B) $x = 60 \text{ cm}$ (C) $y = 75 \text{ cm}$ (D) $y = 130 \text{ cm}$
49. A metallic wire of length L is held between two rigid supports with no initial stress. If the wire is cooled through a temperature θ , then fundamental frequency of oscillation is proportional to :
- (where, Y = Young's modulus of elasticity of wire, ρ = density, α = thermal coefficient of linear expansion)
- (A) $\frac{1}{L}$ (B) $\sqrt{Y}$ (C) $\sqrt{\alpha/\rho}$ (D) $\sqrt{\theta}$
50. A spring of spring constant K is fixed to the ceiling of a lift. The other end of the spring is attached to a block of mass m . The mass is in equilibrium. Now the lift accelerates downwards with an acceleration $2g$. Then choose the correct option(s).
- (A) The block will not perform SHM and it will stick to the ceiling
- (B) The block will perform SHM with time period $2\pi\sqrt{m/K}$
- (C) The amplitude of the block will be $2mg/K$ if it performs SHM
- (D) The minimum potential energy of the spring during the motion of the block will be 0
51. A certain pure metal has a specific heat of $230 \text{ J kg}^{-1} \text{ K}^{-1}$ at high temperatures. Which of these are correct?
- (A) Molar specific heat of metal is 24.942 J/mol-K
- (B) Molar specific heat of metal remains constant for a slight temperature change
- (C) Molecular weight of metal is 108 gram
- (D) All atoms in metal vibrate at same frequency
52. A thin cylindrical metal rod is bent into a ring with a small gap as shown in figure. On heating the system:
- (A) θ decreases, r and d increases (B) r and d increases
- (C) θ , d and r increases (D) θ is constant

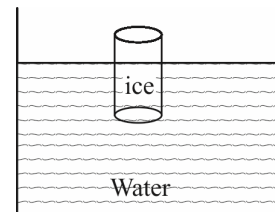


53. A mercury barometer is used to measure air pressure in an elevator. If the elevator is stationary, the reading of the barometer is 76 cm. Assume air pressure to remain unchanged, choose correct statements.
- (A) If the elevator is accelerated upwards, the reading of barometer is more than 76 cm
 (B) If the elevator is accelerated upwards, the reading of barometer is less than 76 cm
 (C) If the elevator is moving upwards with constant velocity, the reading of barometer is 76 cm
 (D) If the elevator is accelerated downwards with acceleration less than g , the reading of barometer is more than 76 cm
54. A light container having a diatomic gas enclosed within, is moving with velocity v . Mass of gas is m , number of moles is n . Then, choose the correct options.
- (A) KE of gas with respect to COM of system is $\frac{3}{2}nRT$
 (B) KE of gas with respect to COM of system is $\frac{5}{2}nRT$
 (C) KE of gas with respect to ground is $\frac{5}{2}nRT$
 (D) KE of gas with respect to ground is greater than $\frac{5}{2}nRT$
55. Two sounding bodies are producing progressive sound wave given by $y_1 = 4\sin(400\pi t)$ and $y_2 = 3\sin(404\pi t)$, where t is in second. The two waves superpose near the ears of a person. The person will hear:
- (A) 2 beats per second
 (B) intensity ratio 7 between maxima and minima
 (C) intensity ratio 49 between maxima and minima
 (D) 4 beats per second
56. During melting of a slab of ice at 273 K and at atmospheric pressure, which of these are correct ?
- (A) positive work is done on ice-water system by atmosphere
 (B) positive work is done on atmosphere by ice-water system
 (C) internal energy of the ice-water system increases
 (D) internal energy of atmosphere increases
57. Initially spring is compressed by x_0 and blocks are in contact. When system is released, then blocks starts moving and after some time contact between blocks breaks, then:



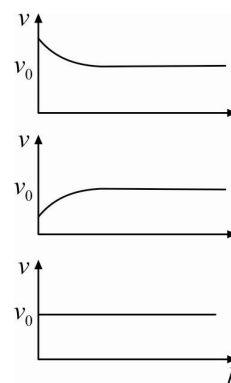
- (A) blocks will separate at natural length of spring
 (B) after separation, block A performs SHM of amplitude $x_0 \sqrt{\frac{M_1}{M_1 + M_2}}$
 (C) after separation, maximum velocity of block A is $x_0 \sqrt{\frac{k}{M_1 + M_2}}$
 (D) after separation, block A will perform SHM of amplitude x_0

58. A cylindrical ice block is floating in water. 10% of its total volume is outside water. Kerosene oil (relative density = 0.8) is poured slowly on top of water in the container. Assume that the oil does not mix with water. Height of the ice cylinder is H . Neglect any melting of ice. Choose the **CORRECT** option(s) from the following.



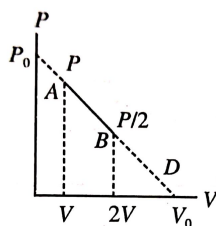
- (A) As kerosene is poured, the volume of ice block above the water level increases and then becomes constant
 (B) As kerosene is poured, the volume of ice block above the water level decreases and then becomes constant
 (C) Thickness of kerosene layer above the water, when 20% of the volume of the ice block is above the water surface, is $\frac{H}{8}$
 (D) Thickness of kerosene layer above the water, when 20% of the volume of the ice block is above the water surface, is $\frac{H}{6}$

59. A student drops several small identical stones from different heights over a deep lake and prepares graphs between speed v of every stone in the water and time t . The graphs can be divided into three categories as shown here. Which of the following explanations of these graphs appear reasonable?



- (a) The first graph is for a stone dropped from a small height, and the second is for a stone dropped from a large height.
 (b) The first graph is for a stone dropped from a large height, and the second is for a stone dropped from a small height.
 (c) The third graph is for a stone dropped from a height sufficient to acquire the speed v_0 at the instant it enters the water.
 (d) The third graph is not possible.

60. An ideal gas is taken from the state A (pressure P , volume V) to the state B (pressure $P/2$, volume $2V$) along a straight line path in the P - V diagram. Select the correct statement(s) from the following :

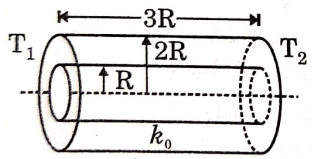
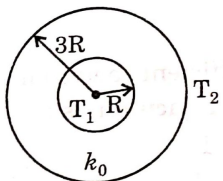
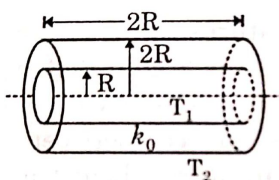
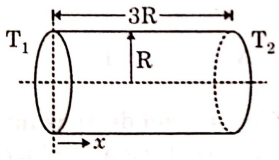


- (A) The work done by the gas in process AB is greater than the work that would be done if the system were taken from A to B along the isotherm
 (B) In the T - V diagram, the path AB becomes a part of parabola
 (C) In the P - T diagram, the path AB becomes a part of hyperbola
 (D) In going from A to B, the temperature T of the gas first increases to a maximum value and then decreases

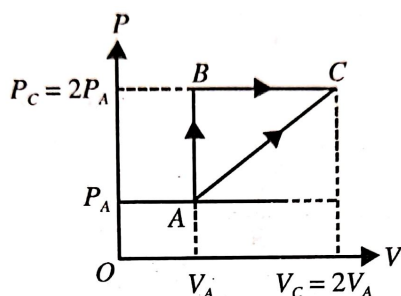
SECTION-4

COLUMN MATCHING TYPE

61. Entries in column I consists of diagrams of thermal conductors. The type of conductor and direction of heat flows are listed below. Entries in column II consists of the magnitude of rate of heat flow belonging to any of the entries in column I. If temperature difference in all the cases is $(T_1 - T_2)$ then match the column :

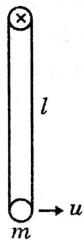
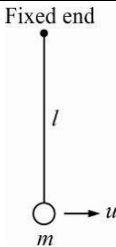
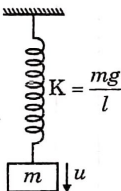
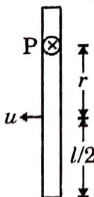
	Column I		Column II
(A)	 Thick cylindrical shell, flow along axis	(p)	$6\pi k_0 R (T_1 - T_2)$
(B)	 Thick spherical shell, radial flow	(q)	$\frac{\pi k_0 R}{2 \ln 2} (T_1 - T_2)$
(C)	 Thick cylindrical shell, radial flow	(r)	$\pi k_0 R (T_1 - T_2)$
(D)	 Solid cylinder, flow along axis, variable k as $k = k_0 \left(1 + \frac{x}{(3R)} \right)$	(s)	$\frac{4\pi k_0 R}{\ln 2} (T_1 - T_2)$

62. n kmol of a monoatomic ideal gas is taken quasi-statically from state A to state C along the straight line shown in figure. Alternatively, the same gas is taken quasi-statically from A to C along the path ABC. Express all answers in terms of P_A and V_A .

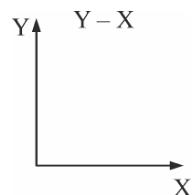


	Column I		Column II
(A)	The heat ΔH added to the gas along the straight line path AC	(p)	$2P_A V_A$
(B)	Change in internal energy of the gas along the straight line path AC	(q)	$\frac{13}{2} P_A V_A$
(C)	The work done by the gas along the path ABC	(r)	$6P_A V_A$
(D)	The heat $\Delta H'$ added to the gas along the path ABC	(s)	$\frac{9}{2} P_A V_A$

63. MATCH THE COLUMN :

	Column I		Column II
(A)	A particle of mass m joined to one end of a light rod of length l where other end is fixed and system is free to rotate in vertical plane about fixed and smooth axis. The particle is given a horizontal velocity u 	(p)	Time period for oscillation can be $2\pi\sqrt{\frac{l}{g}}$
(B)	A particle of mass m attached to one end of a light string of length l which can rotate in the vertical plane about other fixed end. The particle is given a horizontal velocity u 	(q)	If $u = \sqrt{4.5gl}$ spring or rod may experience compression force during the motion or string may get slack at some stage in motion.
(C)	A block of mass m attached to one end of an ideal spring of force constant K and free to oscillate in the vertical direction. The block initially hangs in equilibrium. It is given a downward velocity u 	(r)	If $u = \sqrt{gl}$ time period for oscillations may be $2\pi\sqrt{\frac{l}{g}}$.
(D)	A uniform rod of mass m and length l free to rotate in the vertical plane about a smooth hinged point P at a distance r from its centre of mass. Value of r can be adjusted. The centre is given a horizontal velocity u 	(s)	Total mechanical energy will be conserved during the motion assuming no air resistance.
		(t)	Angular amplitude of oscillations may be $\frac{2\pi}{3}$

64. For an adiabatic process, match the graphs given in column II corresponding to the pair given in column I taking the first quantity on Y-axis and second quantity on X-axis.

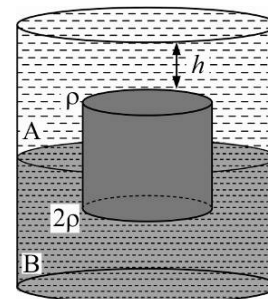


	Column I		Column II
(A)	Pressure-Temperature	(P)	
(B)	Volume-Temperature	(Q)	
(C)	Pressure-Volume	(R)	
(D)	Pressure-Internal Energy	(S)	

65. Column I shows different sets of standing waves in a string of length L whose ends are fixed or free according to respective figure and Column-II shows possible equations for them where symbols have usual meaning.

	Column I		Column II
(A)		(P)	$y = A \sin \frac{\pi x}{L} \cos \omega t$
(B)		(Q)	$y = A \sin \frac{2\pi x}{L} \cos \omega t$
(C)		(R)	$y = A \cos \frac{\pi x}{2L} \cos \omega t$
(D)		(S)	$y = A \cos \frac{\pi x}{L} \cos \omega t$

66. Shown below is a vessel containing liquids A and B of densities ρ and 2ρ respectively. A cylinder of height $2h$ floats with equal volumes submerged in both liquids. The top surface of cylinder is at a depth h below the surface of liquid A . Neglecting atmospheric pressure, match the quantities mentioned in column I with corresponding expression in column II.



	Column I		Column II
(A)	Net force exerted by liquid A on the cylinder	(P)	$9\rho g R h^2$
(B)	Net force exerted by liquid B on the cylinder	(Q)	$\pi\rho g R^2 h$
(C)	Net force exerted by liquids A and B on the left half of the curved part of cylinder	(R)	$4\pi\rho g R^2 h$
(D)	Net force exerted by liquids A and B on the cylinder	(S)	$3\pi\rho g R^2 h$
		(T)	$2\pi\rho g R^2 h$

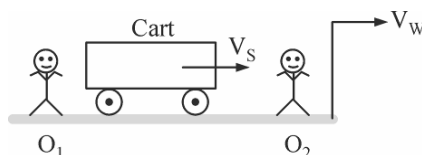
67. Two particle 'A' and 'B' start SHM at $t = 0$. Their positions as a function of time are given by:

$$x_A = A \sin \omega t$$

$$x_B = A \sin(\omega t + \pi/3)$$

	Column I		Column II
(A)	Minimum time when x is same	(P)	$\frac{5\pi}{6\omega}$
(B)	Minimum time when velocity is same	(Q)	$\frac{\pi}{3\omega}$
(C)	Minimum time after which $v_A < 0$ and $v_B < 0$	(R)	$\frac{\pi}{\omega}$
(D)	Minimum time after which $x_A < 0$ and $x_B < 0$	(S)	$\frac{\pi}{2\omega}$

68. Observers O_1 and O_2 are at rest and the wall is moving with velocity V_w . Cart is moving with constant velocity V_s towards wall. The source of sound is in the cart, the original frequency of the wave is f . Sound has velocity C w.r.t. ground (medium is stationary). Then match the column-I with column II.



	Column I		Column II
(A)	Wavelength received by O_1 directly from cart	(P)	$\left(\frac{C-V_w}{C+V_w}\right)\frac{(C-V_s)}{f}$
(B)	Wavelength received by O_2 directly from cart	(Q)	$\frac{C+V_s}{f}$
(C)	Wavelength received by driver of the cart after reflection from wall.	(R)	$\left(\frac{C+V_w}{C-V_w}\right)\frac{(C-V_s)}{f}$
(D)	Wavelength received by O_1 after reflection from wall	(S)	$\frac{C-V_s}{f}$
		(T)	None of these

69. In the table shown below, Column-II shows the possible outcomes to the water level of a swimming pool when a person standing on a boat in it does any one of actions shown in Column-I. Match the possible outcomes.

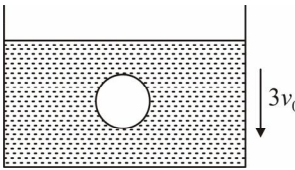
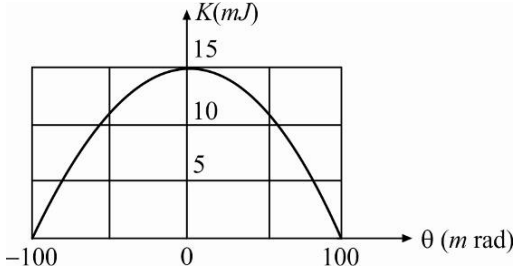
	Column I		Column II
(A)	He throws a 20 kg iron anchor from the boat into water, which the settles at the bottom	(P)	It becomes lower
(B)	He throws out a 20 kg log from the boat. The log floats on water	(Q)	It becomes higher
(C)	He empties 20 kg of water from the boat into the pool	(R)	It stays the same
		(S)	Cannot be predict from the information given

70. A uniform wire of length l , cross sectional area A and Young's modulus Y is subjected to deforming forces in different ways mentioned in column I and it effects certain parameters which depend on variables given in Column II.

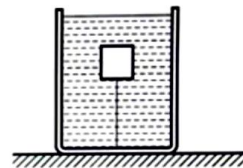
	Column I		Column II
(A)	If the wire is pulled at its ends by equal and opposite forces of magnitude F so that it undergoes an elongation x , according to Hooke's law, $F = kx$, where (k) of the wire will depend on	(p)	Young's modulus Y
(B)	Let us suspend the wire vertically from a rigid support and attach a mass m at its lower end. If the mass is slightly pulled down and released, it executes S.H.M. of a time period that will depend on	(q)	Elongation (x)
(C)	If the given wire is fixed between two rigid supports at a separation of l and its temperature is increased, thermal stress that develops in the rod will depend on	(r)	Length (l)
(D)	Work done in stretching the wire to a length $l + x$ will depend on	(s)	Area of cross section (A)

SECTION-5

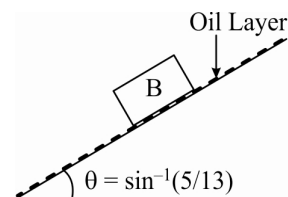
NUMERICAL TYPE

71. A string of length ' l ' is fixed at both ends. It is vibrating in its 3rd overtone with maximum amplitude ' a '. The amplitude at a distance $l/3$ from one end is ' a' '. Find the value of $\frac{a'}{a}$.
72. A container filled with viscous liquid is moving vertically downwards with constant speed $3v_0$. At the instant shown, a sphere of radius r moving vertically downwards, in liquid, has a speed v_0 . The coefficient of viscosity is η . There is no relative motion between the liquid and the container. Then at the shown instant, the magnitude of viscous force acting on sphere is $a \times \pi \eta r v_0$, where a is _____.
- 
73. A small ball is suspended by a thread of length $l = 1\text{m}$ at the point O on the wall, forming a small angle $\alpha = 2^\circ$ with the vertical (as shown in figure). Then the thread with ball was deviated through a small angle $\beta = 4^\circ$ and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period (in sec) of such a pendulum. (Take $g = \pi^2$)
- 
74. A new scale N of temperature is divided in such a way that the freezing point of water is $100^\circ N$ and boiling point is $400^\circ N$. The Celsius and new temperature scale readings will be same at _____ $^\circ C$.
75. The time period of a physical pendulum is given by $T = 2\pi \sqrt{\frac{I}{mgl}}$ where m = mass of the pendulum, I = moment of inertia about the axis of suspension, l = distance of centre of mass of body from the axis of suspension. The change in time period when temperature changes by $\Delta\theta$ is given by $\frac{\alpha\Delta\theta}{n}T$, then the value of n is _____. [The coefficient of linear expansion of the material of pendulum is α]
76. Figure shows the kinetic energy K of a simple pendulum versus its angle θ from the vertical. The pendulum bob has mass 0.2 kg . If the length of the pendulum is equal to $\frac{x}{2}m$, then value of x is _____. ($g = 10\text{m/s}^2$)
- 
77. A steel ball initially at a pressure of 10^5 Pa is heated from $20^\circ C$ to $120^\circ C$ keeping its volume constant. The final pressure inside the ball is _____ $\times 10^8\text{ Pa}$. (Given that coefficient of linear expansion of steel is $1.1 \times 10^{-5}^\circ C^{-1}$ and bulk modulus of steel is $1.6 \times 10^{11}\text{ N/m}^2$)
78. A cubical block of wood of specific gravity 0.5 and chunk of concrete of specific gravity 2.5 are fastened together. The ratio of mass of wood to the mass of concrete which makes the combination to float with entire volume of the combination submerged in water is _____.

79. Figure shows a solid block of volume 1000 cm^3 and density 0.8 gm/cm^3 submerged in a liquid of density 1.2 gm/cm^3 and tied to bottom with a cord. If container is accelerating up at 5 m/s^2 then find tension in cord (in N).

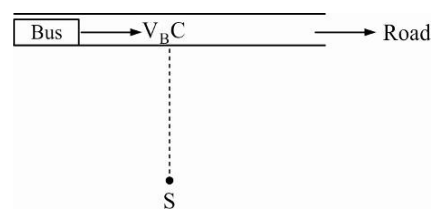


80. A block B of mass 13 kg and base area $1\text{m} \times 1\text{m}$ is sliding on an inclined plane lubricated by an oil layer of thickness 0.15 cm . It is observed that block slides with a steady speed of 0.5 m/s . Find viscosity of oil in SI units. (Take $g = 10 \text{ m/s}^2$)



81. A thin uniform rod of length $2l$ and specific gravity 0.75 is hinged at one end to a point $l/2$ above the water surface with other end immersed in water. The rod inclination to horizontal in equilibrium is $\frac{\pi}{n}$ where n is _____.

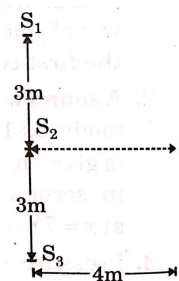
82. A bus is moving with a velocity v_B along a road as shown in the figure. A shooter S is at a distance l from the road. He has a detector which can detect signals only of frequency 1500 Hz . The bus is blowing horn of frequency 1000 Hz . When detector detects a signal, shooter immediately shoots towards the road along SC and bullet hits the bus. If



velocity of sound in medium is 360 m/s and $\frac{v_{bus}}{v_{sound}} = \frac{2}{3\sqrt{3}}$,

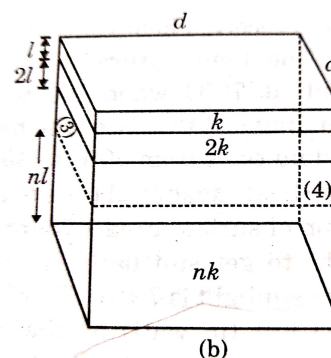
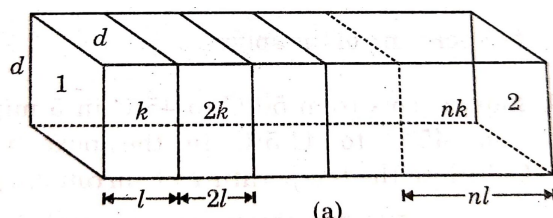
then velocity of bullet (in m/s) is _____.

83. The three identical loud speakers in figure play a 170 Hz frequency tone in a room where speed of sound is 340 m/s . At point exactly in front of source S_2 the amplitude of the wave from each speaker is a . What is ratio of resultant intensity at P and intensity due to single speaker ?



84. At a particular instant a source of sound of frequency 100 Hz is at $(2\text{m}, 1\text{m})$ and an observer is at $(5\text{m}, 5\text{m})$. The velocity of source and observer are $15(2\hat{i} + \hat{j}) \text{ m/s}$ and $5\hat{i} + 15\hat{j} \text{ m/s}$ respectively at this instant. The velocity of sound in air is 330 m/s . The frequency of the sound received by the observer is found to be $15x \text{ Hz}$. Find the value of x .

85. An arrangement consisting of slabs is shown in the figure. When system is placed in situation (a) and temperature difference is maintained between (1) and (2), equivalent thermal conductivity of system is K and when system is changed to situation (b) and temperature difference is maintained between (3) and (4) then equivalent thermal conductivity increases by 20% . Find number of slabs in the arrangement.



SECTION-6

SUBJECTIVE TYPE

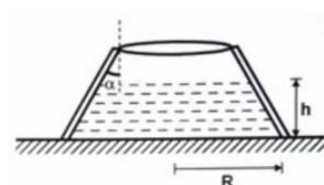
86. A piston can freely move inside a horizontal cylinder closed from both ends. Initially, the piston separates the inside space of cylinder in two equal parts each of volume V_0 which are filled with an ideal gas at same pressure P_0 . If temperature of gas in two parts is equal and kept constant, find the work required to be done in increasing the volume of one part of gas η times compared to the other part by slowly displacing the piston.

87. In a given force field, the potential energy of a particle is given as a function of its x -coordinate as

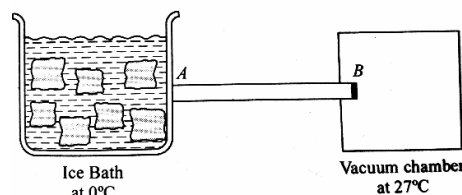
$$U_{(x)} = \frac{p}{x^2} - \frac{q}{x}$$

Where p and q are positive constants. Find the period of small oscillations of the particle about its equilibrium position in the field.

88. A conical vessel without a bottom stands on a smooth table. A liquid is poured in the vessel and as soon as liquid level rises to h , the vessel is lifted up at the bottom by liquid pressure. The base radius of vessel is R and its weight is W . Find the density of liquid.

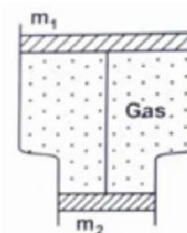


89. A cylindrical rod of 50 cm length and having 1 cm^2 cross sectional area is used as a conducting material between an ice bath at 0°C and a vacuum chamber at 27°C as shown in figure. The end of rod which is inside the vacuum chamber behaves like a black body and is at temperature 17°C in steady state. Find the thermal conductivity of the material of rod and rate at which ice is melting in the ice bath. Given that latent heat of fusion of ice is $3.35 \times 10^5 \text{ J/kg}$.

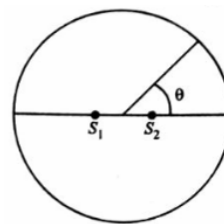


90. Three rods A , B and C having identical shape and size, are hinged together at ends to form an equilateral triangle. Rods A and B are made of same material having coefficient of linear thermal expansion α_1 while that of material of rod C is α_2 . By what temperature should the system of rods be heated to increase the angle opposite to rod C by $\Delta\theta$.

91. A smooth vertical tube having two different cross sections is open from both the ends but closed by two sliding pistons as shown in figure. Pistons are tied with an inextensible string. One mole of an ideal gas is enclosed between the pistons. The difference in cross-sectional areas of pistons is ΔS and masses of pistons are m_1 and m_2 . Find the temperature by which tube is raised so that pistons will be displaced by a length ℓ . Take atmospheric pressure to be P_0 .



92. Figure shows two coherent sources S_1 and S_2 which emit sound of wavelength λ in phase. The separation between the sources is 3λ . A circular wire of large radius is placed in such a way that S_1S_2 lies in its plane and the middle point of S_1S_2 is at the centre of the wire. Find the angular positions θ on the wire for which constructive interference takes place.

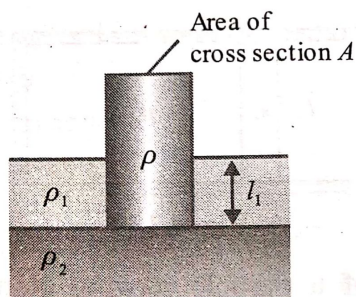


93. A soap bubble of radius r and surface tension T is given some charge and as a result of which its potential becomes V volt. Show that the new radius R of soap bubble is given by the equation

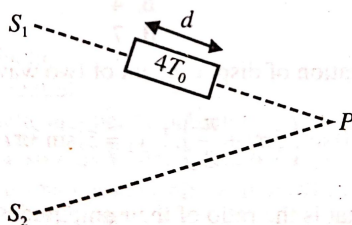
$$P_0(R^3 - r^3) + 4T(R^2 - r^2) - \frac{\epsilon_0 V^2 R}{2} = 0$$

where P_0 is the atmospheric pressure.

94. A cubical container of side a and wall thickness x ($x \ll a$) is suspended in air and filled with n moles of a diatomic gas (adiabatic exponent $= \gamma$) in a room where room temperature is T_0 . If at $t = 0$ gas temperature is T_1 ($T_1 > T_0$), find the temperature as a function of time t . Assume the heat is conducted through all the walls of container.
95. A vertical pole of length l , density ρ , area of cross section A floats in two immiscible liquids of densities ρ_1 and ρ_2 . In equilibrium position the bottom end is at the interface of the liquids. When the cylinder is displaced vertically, find its time period of oscillation.



96. Sound from two coherent sources S_1 and S_2 are sent in phase and detected at point P equidistant from both the sources. Speed of sound in normal air is V_0 , but in some part in path S_1P , there is a zone of hot air having temperature 4 times of the normal temperature, and width d . What should be minimum frequency of sound, so that minima can be found at P ?



97. Consider a cubical vessel of edge a having a small hole in one of its walls. The total thermal resistance of the walls is r . At time $t = 0$, it contains air at atmospheric pressure p_a and temperature T_0 . The temperature of the surrounding air is T_a ($T_a > T_0$). Find the amount of the gas (in moles) in the vessel at time t . Take C_v of air to be $5R/2$.

98. A horizontal tube of length l closed at both ends contains an ideal gas of molecular weight M . The tube is rotated at a constant angular velocity ω about a vertical axis passing through an end. Assuming the temperature to be uniform and constant, show that $p_2 = p_1 e^{\frac{M\omega^2 l^2}{2RT}}$, where p_2 and p_1 denote the pressures at the free end and the fixed end respectively.
99. A cylindrical tank of base area A has a small hole of area a at the bottom. At time $t = 0$, a tap starts to supply water into the tank at a constant rate $\alpha \text{ m}^3/\text{s}$.
- what is the maximum level of water $h_{\max}$ in the tank ?
 - find the time when level of water becomes $h (< h_{\max})$.
100. A bubble of radius r , containing a diatomic ideal gas, has the soap film of thickness h and is placed in vacuum. The soap film has surface tension σ and density ρ .
- Find the molar heat capacity of the gas in the bubble for such a process when the gas is heated so slowly that the bubble remains in a mechanical equilibrium.
 - Find the angular frequency ω of the small radial oscillations of the bubble under the assumption that the heat capacity of the soap film is much greater than the heat capacity of the gas in the bubble. Assume that the thermal equilibrium inside the bubble is reached much faster than the period of oscillations.